



Research Paper

VMGP: A unified variational auto-encoder based multi-task model for multi-phenotype, multi-environment, and cross-population genomic selection in plants

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ABSTRACT

Plant breeding stands as a cornerstone for agricultural productivity and the safeguarding of food security. The advent of Genomic Selection heralds a new epoch in breeding, characterized by its capacity to harness whole-genome variation for genomic prediction. This approach transcends the need for prior knowledge of genes associated with specific traits. Nonetheless, the vast dimensionality of genomic data juxtaposed with the relatively limited number of phenotypic samples often leads to the “curse of dimensionality”, where traditional statistical, machine learning, and deep learning methods are prone to overfitting and suboptimal predictive performance. To surmount this challenge, we introduce a unified Variational auto-encoder based Multi-task Genomic Prediction model (VMGP) that integrates self-supervised genomic compression and reconstruction with multiple prediction tasks. This approach provides a robust solution, offering a formidable predictive framework that has been rigorously validated across public datasets for wheat, rice, and maize. Our model demonstrates exceptional capabilities in multi-phenotype and multi-environment genomic prediction, successfully navigating the complexities of cross-population genomic selection and underscoring its unique strengths and utility. Furthermore, by integrating VMGP with model interpretability, we can effectively triage relevant single nucleotide polymorphisms, thereby enhancing prediction performance and proposing potential cost-effective genotyping solutions. The VMGP framework, with its simplicity, stable predictive prowess, and open-source code, is exceptionally well-suited for broad dissemination within plant breeding programs. It is particularly advantageous for breeders who prioritize phenotype prediction yet may not possess extensive knowledge in deep learning or proficiency in parameter tuning.

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1. Introduction

The imperative to bolster global food security is more pressing than ever, with the specter of a burgeoning population projected to reach 9.7 billion by 2050 (Gebbers and Adamchuk, 2010; Tilman et al., 2011; Ehrlich and Harte, 2015). This demographic surge is compounded by a confluence of environmental and societal shifts, including land degradation, water scarcity, climate volatility, and evolving sociocultural dynamics (Kamilaris et al., 2017; Harfouche et al., 2019). In this context,

plant breeding emerges as a linchpin strategy, one that must navigate the multifaceted demands of enhancing yield, quality, and stress resilience (Hallauer, 2007; Tester and Langridge, 2010; Varshney et al., 2020).

Historically, the utilization of phenotypic, pedigree, and performance data has catalyzed significant enhancements in plant varieties throughout the 20th century (He and Li, 2020). Given that traits under selective breeding are often governed by quantitative trait loci (QTLs), genomic selection (GS) has risen to prominence as an innovative approach. This method harnesses the wealth of whole-genome variations to construct genomic prediction models, circumventing the need for prior identification of trait-associated genes. It promises a transformative impact on plant breeding by abbreviating breeding cycles and

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amplifying the precision of selection (Desta and Ortiz, 2014; Crossa et al., 2017; Wallace et al., 2018; Cui et al., 2020; Yan et al., 2021; Zhao et al., 2022; Casale et al., 2022; Li et al., 2024; Wang et al., 2025).

The genesis of GS is rooted in its application within animal husbandry, where it was pivotal in estimating the breeding values of unproven bovine specimens (Meuwissen et al., 2001). This pioneering application has since spawned a multitude of statistical and computational methodologies tailored for GS. These include genomic best linear unbiased prediction (GBLUP), ridge regression-best linear unbiased prediction (rrBLUP), a spectrum of Bayesian algorithms such as BayesA, BayesB, and BayesCπ, and machine learning algorithms, including random forest (RF), support vector regression (SVR), and the light gradient boosting machine (LightGBM) (Meuwissen et al., 2001; Park and Casella, 2008; Heffner et al., 2010; Endelman, 2011; Habier et al., 2011; Heslot et al., 2012; Montesinos-López et al., 2019; Yan et al., 2021). These methodologies have been successfully operationalized in numerous plant and animal breeding initiatives, yielding substantial genetic advancements.

Deep learning (DL), as the best solution in the fields of natural language processing, computer vision, and multi-modal learning (LeCun et al., 2015; Tsaftaris et al., 2016; Mohanty et al., 2016; Tang et al., 2023), has been introduced to the realm of GS with the aim of refining genotype-phenotype predictions. DL-based GS methods typically use deep neural networks (e.g., fully connected layers or convolutional layers) to process genomic data along with nonlinear activation functions and other techniques to elucidate the additive, dominant, and epistatic effects within genomic data. These models, theoretically, hold the potential to surpass the predictive prowess of traditional statistical approaches such as mixed linear models. However, empirical findings suggest that DL models may only match the performance of GBLUP or rrBLUP in practical scenarios (Ma et al., 2018; Montesinos-López et al., 2021; Nazzicari and Biscarini, 2022).

In the quintessential GS task, the genotypes and phenotypes of a reference population (training set) are meticulously measured, and a GS model is trained on this data to delineate the genotype-phenotype correlation within the population. Subsequently, the phenotypes of a candidate population (testing set) can be inferred by the trained model using genotypes as inputs (Jiang et al., 2020). The input features' dimensionality is notably high, often encompassing tens of thousands to millions of single nucleotide polymorphisms (SNPs), contingent upon the sequencing techniques employed (Yan et al., 2021). Conversely, the training population's sample size is typically modest, ranging from hundreds to thousands, due to the protracted time frame and prohibitive costs associated with phenotype collection. This “curse of dimensionality” (CoD) presents a formidable challenge in GS, where traditional statistical or machine learning methods, and even DL models, are prone to overfitting, culminating in suboptimal generalization and predictive performance (Zhang et al., 2021).

Auto-encoders (AEs), leveraging the self-supervised learning capabilities inherent to genomic data, have demonstrated efficacy in the medical field for addressing the CoD. Variants such as reduced noise auto-encoders and variational auto-encoders (VAEs) have been instrumental in cancer genomic data analysis, enabling dimensionality reduction, tumor classification, and prognostic forecasting (Way and Greene, 2018; Qiu et al., 2020; Zhang et al., 2021; Arafa et al., 2023). Moreover, AE-based methodologies have been proposed for single-cell multi-omics analysis, addressing a spectrum of tasks including embedding, clustering, batch correction, imputation, and visualization (Amodio et al., 2019; Gala et al., 2021; Kalafut et al., 2023; Tang et al., 2023). AEs offer a distinct advantage over traditional dimensionality reduction techniques, such as principal component analysis, by revealing the underlying data structure without the necessity for predefined similarity or distance metrics in the original data space (Amodio et al., 2019; Zhang et al., 2021). The success of these studies underscores the potential of AEs in processing genomic data. The VAE architecture extends conventional AEs through its probabilistic framework, employing a

Gaussian-constrained latent space and reparameterized sampling mechanism. This design enables effective modeling of novel genetic variants encountered in GS contexts while inherently mitigating overfitting risks through controlled stochasticity in latent representations. These key attributes that motivated its integration as the foundational component of this study. However, AEs' application has been predominantly confined to single-cell multi-omics research, with limited forays into individual or population-level genomics, particularly within the scope of plant genomic selection.

The genotype-phenotype relationship is characterized by substantial variability, with inconsistent distributions across different populations. Consequently, GS models are often trained within specific populations for individual phenotypic traits in isolation (Yan et al., 2021; Wang et al., 2023). The challenge of constructing an integrated model that transcends populations and encompasses multiple phenotypes is non-trivial. Genetic pleiotropy, where a single gene influences multiple traits due to its involvement in diverse biosynthetic or metabolic pathways, is a prevalent phenomenon (Watanabe et al., 2019). Cross-population genetic coherence emerges from conserved linkage disequilibrium patterns, shared gene structures and functions, analogous expression regulation, polygenic inheritance, and similar three-dimensional genome and epigenetic architectures (Oleksiak et al., 2002; Ruan et al., 2022; Liu et al., 2023). These factors substantiate the biological plausibility of our multi-task learning framework for GS, that is a unified modeling architecture that simultaneously incorporates multi-phenotype correlations, genotype-environment interactions, and cross-population genetic architectures to capitalize on shared biological signals through synergistic integration of complementary data dimensions. A multi-layer deep learning network can harness shared network layers to extract commonalities across tasks, while task-specific layers can be tailored to address the unique characteristics of various traits, the interplay between genotypes and environments, and the differential expression among populations.

To tackle these challenges, we have engineered a unified Variational auto-encoder based Multi-task Genomic Prediction (VMGP) model, specifically tailored for plant genomic selection (GS). The VMGP model distinguishes itself by adeptly learning the low-dimensional latent representation of genomic data. It is accomplished through self-supervised learning mechanisms that involve the compression and subsequent reconstruction of data, as facilitated by the VAE framework. This process effectively encapsulates the intrinsic information embedded within the genomic sequences. Leveraging this latent genotype representation, in conjunction with a modest collection of multi-dimensional phenotypic data, the VMGP model can be trained to operate an integrated multi-task phenotype prediction framework. This holistic approach not only permits an in-depth exploration of the complex interplay between genotypes and phenotypes but also synchronizes the correlations spanning multiple predictive tasks. Additionally, it augments the sample size of the shared network for individual predictive tasks, thereby significantly enhancing the robustness and reliability of the model.

The VMGP model has demonstrated superior performance over existing state-of-the-art methodologies, including rrBLUP (Endelman, 2011), BayesA, RF, deep neural network for genomic prediction (DNNGP) (Wang et al., 2023), and Cropformer (Wang et al., 2025), across a diverse array of public datasets for maize, rice, and wheat on multi-phenotype, multi-environment, and cross-population GS tasks. This comparative excellence is achieved within a unified framework with consistent parameterization, underscoring the VMGP model's versatility and adaptability.

The superior predictive accuracy of VMGP is a testament to its proficiency in delineating the intricate relationship between genotype and phenotype. The model's user-friendly design and universal applicability are particularly noteworthy, as it delivers robust predictive outcomes without the need for the structural redesign and parameter fine-tuning typically required in conventional deep learning tasks.

2. Materials and methods

2.1. Datasets

This study leveraged four comprehensive datasets spanning three species and encompassing a spectrum of population sizes: rice299, wheat599, wheat2000, and maize4505.

The rice299 dataset, procured from the International Rice Research Institute's irrigated rice breeding program (Spindel et al., 2015), encompasses 299 rice samples characterized by 73,147 SNP markers and four phenotypic traits, including plant height (PH), grain length (GrL), grain width (GrW), and yield (YLD).

The wheat599 dataset represents a compilation of 599 historical wheat lines from the International Maize and Wheat Improvement Center (CIMMYT) Global Wheat Program (Wang et al., 2023). Post-quality control measures, each wheat line was profiled using 1279 SNP markers and assessed for yield (YLD) across four distinct agroclimatic regions.

The wheat2000 dataset is composed of 2000 Iranian bread wheat landraces sourced from the CIMMYT wheat gene bank (Wang et al., 2023). These accessions were genotyped with 33,709 SNP markers and evaluated across six phenotypic traits: thousand grain weight (TGW), test weight (TW), grain length (GrL), grain width (GrW), grain hardness (GrH), and grain protein (GrP).

The maize4505 dataset is a collection of 4505 maize recombinant inbred lines derived from 25 sub-populations of the USNAM population (Hung et al., 2012). Each line was genotyped using a dense set of 106,944 SNP markers and examined for five phenotypic traits: days to pollen (DTP), days to silking (DTS), ear height (EH), hundred kernel weight (HGW), and plant height (PH).

The rice299, wheat2000, and maize4505 datasets were engaged in multi-phenotype prediction tasks, whereas the wheat599 dataset was

specifically utilized for multi-environment prediction tasks. Owing to the maize4505 dataset's detailed sub-population stratification, it was uniquely suited for cross-population prediction tasks. To navigate this intricate scenario, we undertook supplementary explainable learning initiatives for feature relevance analysis and SNP pre-selection within this dataset.

2.2. VMGP: A unified variational auto-encoder based multi-task genomic prediction model

In this study, we introduce the VMGP model, a novel framework designed to surmount the challenges outlined in the Introduction by integrating self-supervised genomic reconstruction with multiple predictive tasks in a seamless manner. The VMGP model is architecturally divided into four distinct yet synergistic modules: the Encoder, Sampler, Decoder, and Predictor (Fig. 1), which synergize to offer exceptional predictive prowess across multiple phenotypes, environments, and populations.

The Encoder and Sampler modules are shared components that process the binary-encoded categorical genomic data X (represented as $[0,1,2]$ genotypes for SNP loci) for both genomic reconstruction and multiple predictive tasks. Let n represent the number of crop samples and n_x denote the number of genomic features. The input genomic data X is thus structured as a matrix in $R^{n \times n_x}$ to encapsulate the relationships between samples and features.

The Encoder module is tasked with distilling the high-dimensional genomic data into a more concise representation, which aids in the efficient encapsulation of pivotal features. This distilled representation forms the bedrock for both tasks, providing a significant abstraction of genotype information. The Encoder consists of two fully connected blocks, each comprising a fully connected layer, a batch normalization

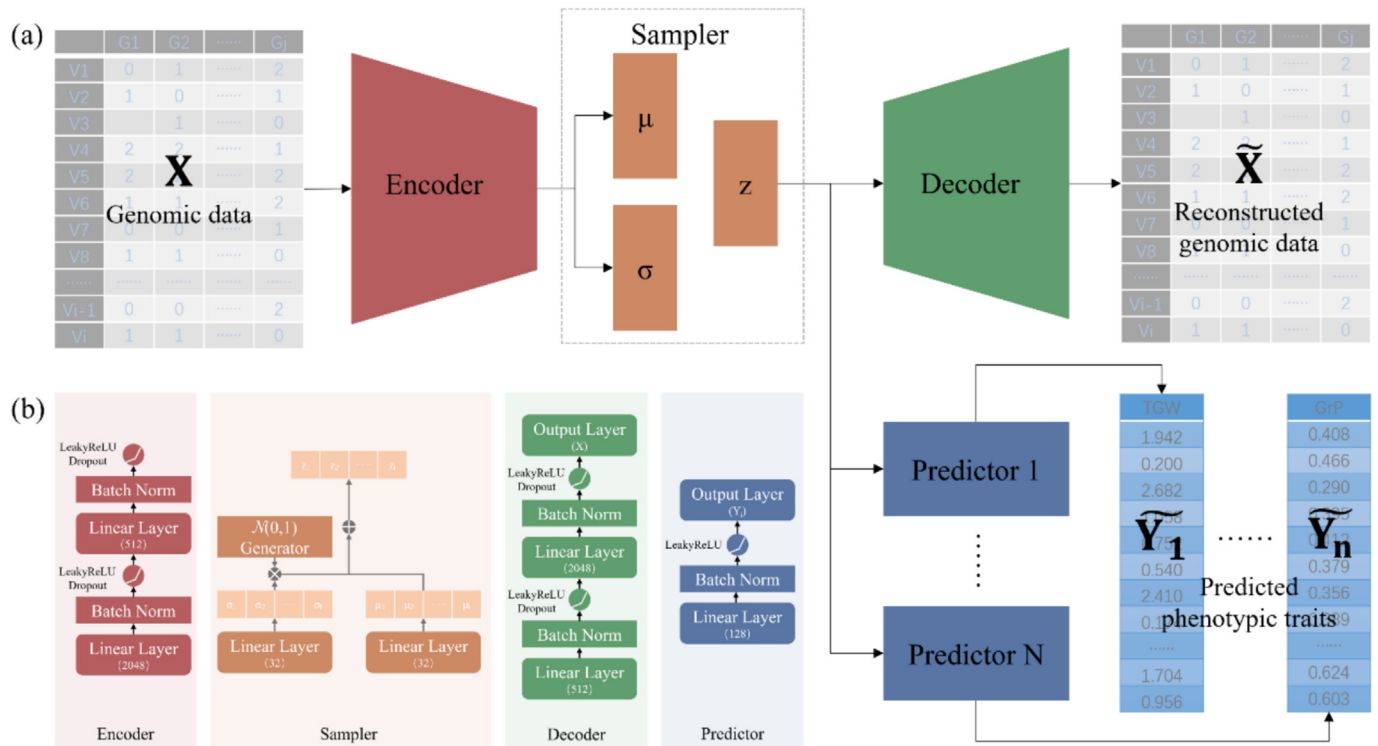


Fig. 1. Framework of the proposed variational auto-encoder based multi-task genomic prediction (VMGP) model. (a) The VMGP model comprises four distinct modules: Encoder, Sampler, Decoder, and Predictor. The genomic data X serves as the input to the model, where it undergoes compression and reconstruction in the Encoder, Sampler and Decoder modules, respectively. The resulting reconstructed genomic data is denoted as $\tilde{X}$. The generated latent genotype embeddings from Sampler are subsequently utilized by Predictor to predict phenotypic traits, which are denoted as $\hat{Y}$. (b) The internal structure diagram of Encoder, Sampler, Decoder, and Predictor.

layer, an activation layer utilizing the Leaky ReLU function,¹ and a dropout layer. These layers are sequentially arranged to transform the feature dimension from n_x to $d1$, and subsequently to $d2$.

The Sampler generates synthetic samples from the encoded representation, employing two bottleneck layers that further compress the encoded data into a mean vector μ and a standard deviation vector σ . These vectors jointly define the Gaussian distribution $\mathcal{N}(\mu, \sigma)$ of the latent variable z given the input sample x . This distribution, denoted as $q(z|x)$, embodies the variational distribution learned by the model. To facilitate gradient backpropagation, the reparameterization trick is implemented, which involves two steps:

- Draw a random variable ε from a standard Gaussian distribution $\mathcal{N}(0, I)$, where I is the identity matrix.
- Compute the sampled variable as:

$$z = \mu + \sigma\varepsilon \quad (1)$$

This reparameterization of the Gaussian distribution enables the Sampler to produce l -dimensional synthetic samples from the encoded representation while allowing for gradient backpropagation during training. This, in turn, enables the model to learn a meaningful representation of the input data and bolster its generalization capabilities.

For the genomic reconstruction task, the output from the Sampler is channeled into the Decoder module. The model then uses the synthetic samples z to reconstruct the genomic data $\tilde{X}$. This reconstruction process ensures that the model captures the essential structure and patterns within the genomic data, providing a solid foundation for accurate predictions. The Decoder consists of two fully connected blocks followed by a linear output layer, arranged inversely to the Encoder, transforming the feature dimension from l to $d2$, then to $d1$, and ultimately to n_x .

In parallel, the predictive tasks diverge by utilizing the encoded representation from the Sampler to feed into their respective Predictors. Each Predictor is customized for a specific predictive task, enabling VMGP to address a variety of predictive challenges from a common low-dimensional genomic expression. Each Predictor is composed of two main components: fully connected blocks sans dropout layer and a linear output layer, designed to transform the feature dimension from l to $d3$, and then to the target output space. The focus of the prediction tasks in this study is the regression of quantitative traits, where the output space typically assumes a scalar value. Thus, each Predictor is configured to produce a single output value corresponding to the predicted trait.

The proposed VMGP model employs a simple and consistent parameter configuration (i.e., 2048, 512, 128, 32 for $d1, d2, d3$ and l , respectively) across diverse GS tasks, including multi-phenotype, multi-environment, and cross-population prediction. This design ensures operational simplicity while remaining highly suitable for widespread promotion and application in plant breeding programs, particularly benefiting breeders prioritizing phenotype prediction accuracy but lacking specialized expertise in deep learning architecture optimization or parameter tuning.

2.3. Loss functions and training process

The VMGP model's joint loss function, akin to its overarching framework, is elegantly composed of two principal components: the

reconstruction task loss and the prediction tasks loss, as delineated in Eq. (2):

$$\mathcal{L} = \alpha \times \mathcal{L}_{rec} + (1 - \alpha) \times \mathcal{L}_{pre} \quad (2)$$

where α is the hyperparameter of loss function adjuster to balance the contribution of two tasks.

The reconstruction loss is a synthesis of two critical elements:

- **Matrix Difference:** This element quantifies the divergence between the reconstructed matrix $\tilde{X}$ and the original genomic data X . Its purpose is to minimize the gap between the original and the reconstructed data, ensuring a precise replication.
- **Kullback-Leibler (KL) Divergence:** This element measures the divergence between two probability distributions. Here, it juxtaposes the inferred distribution of the low-dimensional expression Z against the original data. The KL divergence is instrumental in guaranteeing that the inferred distribution mirrors the original data's fundamental traits while achieving dimensionality reduction.

The formal expression for the reconstruction loss is given in Eq. (3):

$$\mathcal{L}_{rec} = \mathcal{L}_{dif} + \mathcal{L}_{KL} = \text{MSE}(\tilde{X}, X) + D_{KL}(\mathcal{N}(\mu, \sigma) \parallel \mathcal{N}(0, I)) \quad (3)$$

where MSE denotes the mean squared error between matrices, and D_{KL} signifies the KL divergence between distributions. By integrating these components, the reconstruction loss assesses both the fidelity of the reconstructed data to the original and the efficacy of the low-dimensional expression in preserving information, thus striking a balance between precision and data reduction.

The prediction loss aggregates the losses across all prediction tasks. Typically, regression tasks are governed by the MSE loss, while classification tasks are informed by the cross-entropy loss. Given that all prediction tasks in this study pertain to regression, the unified prediction loss is articulated in Eq. (4):

$$\mathcal{L}_{pre} = \sum_i \mathcal{L}_{pre}^{(i)} = \sum_i \text{MSE}(\tilde{Y}_i, Y_i) \quad (4)$$

For the training of the VMGP model, we employed the aforementioned loss function, calibrated for the genomic and phenotypic data of the training samples. Post-training, the genomic data of the test samples are inputted to ascertain their predicted phenotypes. The entire training and prediction regimen is encapsulated in **Algorithm 1 VMGP** (Fig. S1), presenting a methodical and coherent methodology. All experiments adhered to a uniform training parameter set: 200 epochs, a batch size of 32, and a learning rate of 0.0001. For multi-phenotype and multi-environment predictions, a 5-fold cross-validation strategy was implemented. Each dataset was randomly partitioned into 5 segments, each undergoing 5 discrete experiments. In each experiment, 4 segments were allocated for training, while the remaining segment was earmarked for evaluation. For cross-population prediction, an analogous 5-fold cross-validation experiment was executed across sub-populations. The maize4505 dataset was apportioned into 5 folds based on sub-population, with 4 folds designated for model training in each cycle and the remaining fold for evaluation.

2.4. Explainable learning and SNPs pre-selection

SHAP is an explainable learning approach for feature relevance analysis based on Game theory (Lundberg and Lee, 2017; Tang et al., 2023). It offers intuitive explanations and visualizations by quantifying the contribution of each feature to the model's predictive outcomes, thus aiding in the understanding of the reasons behind the model's predictions. The foundation of SHAP lies in the concept of Shapley values, which are utilized to address the issue of contributions in cooperative games. Unlike traditional methods for evaluating feature importance, SHAP offers more comprehensive and accurate information as they

¹ The Leaky ReLU activation function resolves the neuron death phenomenon inherent to standard ReLU by incorporating a learnable negative slope coefficient for inputs below zero, thereby preserving gradient flow during backpropagation. Crucially, this modification retains ReLU's foundational advantages (e.g., computational efficiency and architectural simplicity) while introducing sparsity-aware adaptability, making it particularly effective for high-dimensional sparse genomic feature spaces where rare variant signals require non-zero gradient preservation (LeCun et al., 2015; Zhu et al., 2022).

consider not only the impact of individual features but also their synergistic interactions. This synergistic effect holds significant importance in complex models, as interactions between multiple features can significantly influence predictive outcomes. Using the SHAP method, the contribution of each SNP to the prediction of various phenotypic traits from VMGP can be analyzed as follows:

$$Shapley_j = \sum_{S \in F \setminus \{j\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f_{S \cup \{j\}}(X_{S \cup \{j\}}) - f_S(X_S)] \quad (5)$$

where $Shapley_j$ is the calculated Shapley values for the j th SNP ($j \in 1, \dots, n_x$), F denotes the set of SNPs in X and S denote a subset of F , $|\cdot|$ denotes the number of elements in the set, $F \setminus \{j\}$ stands for dropping the j th feature from the set F , $S \cup \{j\}$ is the union set of S and j , f denotes the predictive function of VMGP. The calculated Shapley values can help gain insights into the extent of each SNP's contribution to the final predictive phenotypes.

Using a pre-selection dataset (ps), which comprises 10 % of the randomly selected samples from the maize4505 dataset, a subset of relevant SNPs (G_{ps}) can be selected by explainable learning for feature relevance analysis. This pre-selection process involves three steps, detailed in **Algorithm 2 SNPs pre-selection** (Fig. S2):

- Train the VMGP net using the pre-selection dataset (X_{train} and Y_{train}).
- Calculate Shapley values for all samples and SNPs for each phenotype, and aggregate Shapley values first by sample and then by phenotype.
- Select SNP the SNP subset based on their aggregated Shapley values.

This pre-selection process not only aids in reducing the dimensionality of the genomic dataset but also enhances the focus on SNPs that contribute significantly to the phenotypes. It can effectively enhance the performance of subsequent prediction tasks, making the selected SNP subset a valuable reference for gene function mining, low-cost sequencing, and other research endeavors.

3. Results

This paper presents the VMGP model without excessive hyperparameter tuning, opting instead for a uniform and standard set of structural parameters (i.e., 2048, 512, 128, 32 for $d1$, $d2$, $d3$ and l , respectively), alongside consistent training parameters (i.e., 200, 32, 0.0001, 0.5 for the number of epochs, batch size, learning rate, and loss function adjuster, respectively). Despite this simplicity, VMGP has demonstrated superior performance across various crop datasets (Figs. 2,3,5), outperforming current state-of-the-art methods.

Our comparative analysis was juxtaposed against five benchmark methods: rrBLUP and BayesA, two preeminent statistical approaches that outperformed machine learning and deep learning techniques in predictive accuracy across numerous studies; RF, a versatile machine learning method underpinned by ensemble learning, renowned for its adaptability and predictive prowess; DNNP and Cropformer, two cutting-edge deep learning methods for genomic selection that garnered impressive predictive results across various datasets.

To ensure equitable comparison, methods not natively designed for multi-task prediction were trained and evaluated independently for each trait or environment. The predictive efficacy was meticulously assessed using two robust metrics: accuracy, as indicated by the Pearson correlation coefficient between the predicted and observed values (termed Accuracy); and prediction error, as measured by the mean absolute error between the predicted and observed values (termed MAE), all within the rigorous framework of a 5-fold cross-validation.

Among the comparative methods, RF and DNNP are sensitive to parameters, hence we conducted a parameter grid search for these methods. The optimal performing parameters (Table S1) and the resultant predictive performance were then utilized as the benchmark results for these comparative methods in subsequent comparisons.

3.1. Multi-phenotype prediction

We conducted an extensive evaluation of the VMGP model for multi-phenotype prediction across three diverse datasets: wheat2000, maize4505, and rice299.

In the wheat2000 dataset, VMGP surpassed all comparative methods in both Accuracy and MAE for all traits. Specifically, VMGP achieved improvements of approximately 4.0 %, 2.0 %, 147.8 %, 9.3 %, and 7.8 % in Accuracy, and reductions of 3.6 %, 2.3 %, 26.6 %, 15.6 %, and 34.4 % in MAE over rrBLUP, BayesA, RF, DNNP, and Cropformer, respectively (Fig. 2a).

Among the comparative methods, BayesA was the most accurate for five traits (TGW, GrW, GrL, GrH, and GrP) and the most precise in MAE for all traits. DNNP secured the highest Accuracy for TW, while RF fell short of expectations, with its performance lagging significantly behind the others. This underscores the divergent predictive capabilities of these methods across different traits. Notably, DNNP's superior performance for certain traits was contingent on trait-specific parameter optimization. In contrast, VMGP's uniform parameterization across all traits, without loss in performance, underscores the model's robust generalization capabilities. While Cropformer demonstrated superior Accuracy among comparative deep learning methods, it exhibited significantly higher MAE values—a phenomenon that might stem from its accuracy-driven early stopping protocol potentially inducing suboptimal convergence within the model's optimization landscape.

To ensure a fair comparison with single-trait prediction methodologies, we introduced a single-trait variant of our model, denoted as VMGP-S, which constructs independent predictive models for each trait. The predictive accuracy and error between VMGP and VMGP-S were found to be nearly identical (Fig. S3), suggesting that the superior performance in phenotype prediction is predominantly attributable to the auxiliary training of the genomic data's compression and reconstruction tasks. This process, facilitated by the VAE framework, effectively extracts informative low-dimensional representations of genotypes, and the joint training of multiple traits does not appear to significantly augment this process.

To further validate this observation, we utilized the VAE framework for pretraining, focusing on the extraction of low-dimensional representations of genomic data, followed by a mere 10 epochs dedicated to fine-tuning the phenotype prediction task, resulting in the VMGP-T model. The performance of VMGP-T was comparable to that of VMGP, with a maximum decrease of 2.8 % in Accuracy and a 6.6 % increase in MAE (Fig. S3). These findings indicate that the low-dimensional expression derived from the self-supervised training of genomic data can efficiently accomplish phenotype prediction tasks, necessitating minimal subsequent adjustment of phenotype-related parameters.

Fig. 2b revealed that VMGP secured the highest average Accuracy and the fourth-best MAE performance in the rice299 dataset. Notably, rrBLUP outperformed VMGP in Accuracy for traits GrL and GrW. RF, which underperformed in the wheat2000 dataset, achieved the lowest average MAE and the fourth-highest average Accuracy. The rice299 dataset exhibits pronounced CoD characteristics, featuring an SNP-to-sample ratio of 245:1. Benchmark analysis revealed conventional statistical approaches (rrBLUP and BayesA) leveraging domain-informed parametric constraints achieved superior accuracy to standard deep learning models (DNNP and Cropformer). VMGP emerged as an exception, attaining state-of-the-art predictive performance through its novel architecture subclass integrating VAE mechanisms (Fig. S9). Moreover, it demonstrated statistically indistinguishable MAE values across five methodologies (excluding Cropformer), with merely 2.9 % range discrepancy between extreme performers. VMGP realized a significant Accuracy excess of 5.7 %, 11.8 %, and 33.9 % over RF, DNNP, and Cropformer, respectively. When focusing on YLD, a paramount comprehensive trait in plant breeding, VMGP's Accuracy exceeded that of rrBLUP, BayesA, RF, DNNP, and Cropformer by 44.2 %, 19.5 %, 24.3 %, 41.4 %, and 82.0 %, respectively.

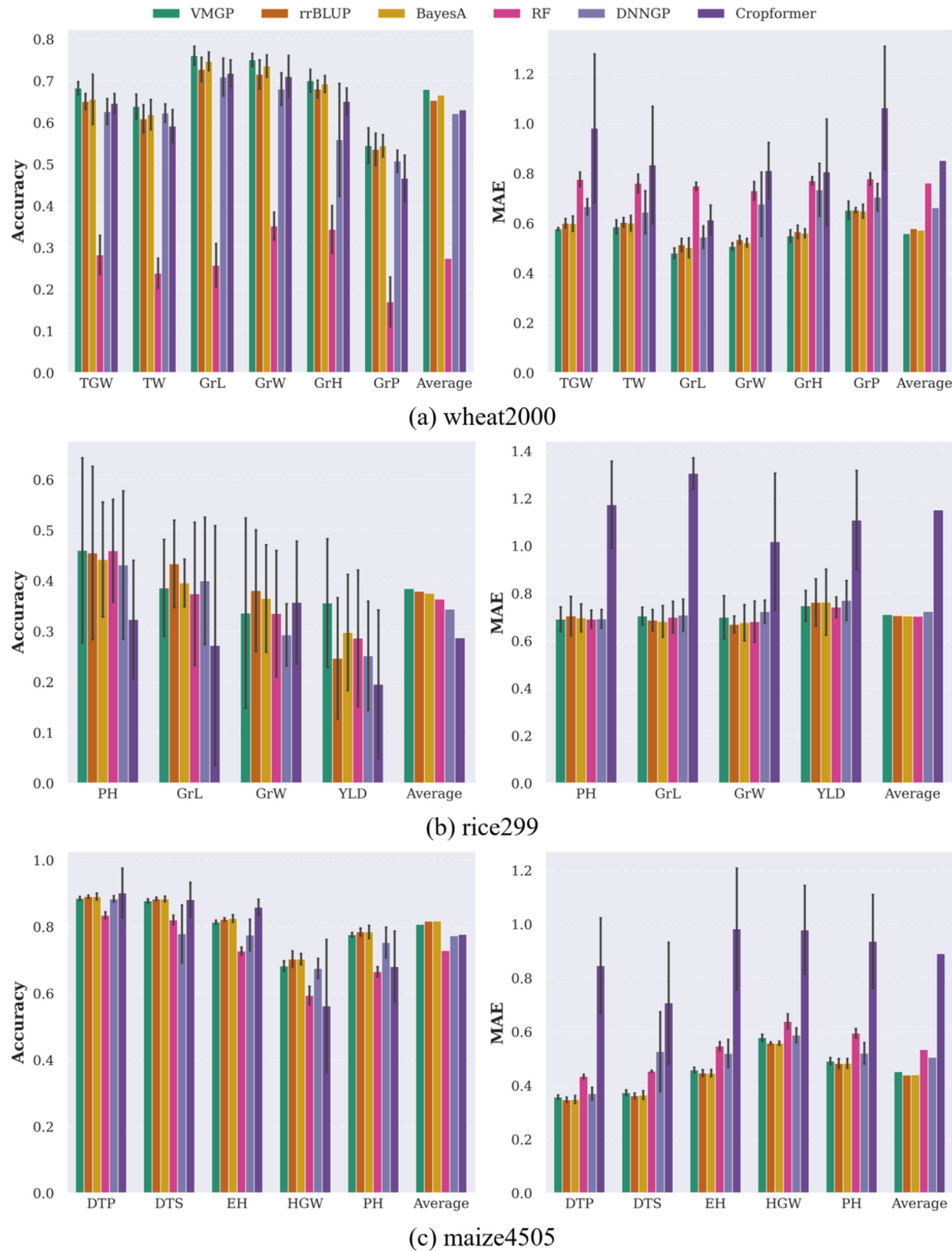


Fig. 2. Multi-phenotype prediction accuracy and prediction error of methods built using four different models on three datasets. (a) wheat2000 dataset. (b) rice299 dataset. (c) maize4505 dataset. VMGP, variational auto-encoder based multi-task genomic prediction; rrBLUP, ridge regression-best linear unbiased prediction; RF, random forest; DNNP, deep neural network for genomic prediction. TGW, thousand grain weight; TW, test weight; GrW, grain width; GrL, grain length; GrH, grain hardness; GrP, grain protein; PH, plant height; YLD, yield; DTP, days to pollen; DTS, days to silking; EH, ear height; and HGW, hundred grain weight.

As illustrated in Fig. S3, the prediction accuracy of the VMGP-T model exhibited a moderate decline of 5.2 %, while the prediction error experienced a relative increment of 2.4 % when compared with the VMGP model. Conversely, the VMGP-S model demonstrated a further enhancement over VMGP, with 1.4 % improvement in Accuracy and 1.1 % reduction in MAE, thereby emerging as the superior method in terms of overall predictive performance. These results imply that the multi-task learning strategy, which integrates VAE-driven genomic reconstruction with single-phenotype prediction, has effectively managed the GS task for the rice299 dataset. The benefits of multi-phenotype joint training in VMGP were primarily observed in the prediction of PH, where it outperformed VMGP-S, indicating that the synergistic training of multiple phenotypes may be particularly advantageous for certain traits.

The maize4505 dataset, showed that rrBLUP and BayesA maintained the best prediction accuracy and error for all traits (Fig. 2c). VMGP was a close third, with a slight 1.3 % decrease in Accuracy and a 2.8 % increase in MAE. VMGP's Accuracy was 10.8 %, 4.4 % and 3.9 % higher than that of RF, DNNGP and Cropformer, while its MAE was 15.3 %, 10.5 % and 49.2 % lower, respectively.

Both VMGP-S and VMGP-T exhibited slightly diminished performance relative to VMGP, with 1.0 % and 1.3 % reductions in Accuracy and 1.2 % and 3.3 % increases in MAE (Fig. S3). Nonetheless, they substantially outperformed RF, DNNGP and Cropformer, reinforcing the efficacy of our VMGP models.

The collective results from these three datasets underscore the variable predictive aptitude of different methods across diverse traits and the necessity of method selection tailored to specific traits. The VMGP model has proven its advantage in predicting a spectrum of traits, positioning it as a versatile GS model for breeders aiming to forecast genotype-phenotype relationships across various crops. It offers invaluable assistance in the predictive profiling of various traits within a diverse array of crop species.

3.2. Multi-environment prediction

A multi-environment prediction experiment was executed utilizing the wheat599 dataset, encompassing yield data across four distinct environments. The absence of direct environmental metrics precluded the explicit modeling of genotype-environment interaction effects. Nonetheless, the phenotypic influence of the environment can be approximated as a composite of fundamental genetic effects and the latent

semantic information specific to each environment, thus permitting an alternative perspective as a special multi-phenotype prediction.

In alignment with the multi-phenotype prediction paradigm, VMGP was employed to conduct multi-environment prediction directly, while comparative methods were trained and assessed in isolation for each environment. The predictive performances were stringently evaluated using Accuracy and MAE within a 5-fold cross-validation framework.

VMGP emerged as the superior model in terms of prediction accuracy and error (Fig. 3). The performance of rrBLUP and RF was closely matched, with a respective decline in prediction accuracy of 7.5 % and 9.3 %, and a modest increase in prediction error of 3.1 % and 4.2 % relative to VMGP. The DNNGP method underperformed relative to the others, while VMGP demonstrated an average Accuracy improvement of 15.7 % and a consequential 8.5 % decrease in error rate.

The single-environment variant of VMGP model, designated as VMGP-S, matched the performance of rrBLUP, a leading comparative method. The incorporation of cross-environment joint training in VMGP resulted in a further enhancement of prediction accuracy by 7.9 % and a reduction in prediction error by 4.3 % over VMGP-S. VMGP-T, despite being subjected to only 10 fine-tuning epochs for yield prediction, surpassed VMGP-S in terms of average prediction accuracy and error rate. Although VMGP-T's prediction error stability was somewhat inferior to that of VMGP due to its abbreviated fitting process for yield values, the benefits of VAE pre-training and multi-environment learning endowed VMGP-T with improved prediction accuracy stability (Fig. S4). The aforementioned results underscore the efficacy of the proposed unified VMGP model for multi-environment genomic prediction, reinforcing its potential for broad application within the field of plant breeding.

3.3. Cross-population prediction

The maize4505 dataset, an aggregation of samples from 20 distinct maize sub-populations, offers a rich ground for cross-population prediction studies. Fig. 4a presents a two-dimensional visualization of the genomic data for these samples, accentuating the varied distribution patterns within each sub-population (McInnes et al., 2018). Fig. 4b provides a quantitative assessment of inter-sub-population genetic divergence through pairwise fixation index (Fst) analysis, with values ranging from 0.230 (2011 vs. 2024) to 0.345 (2006 vs. 2010). Furthermore, the phenotypic diversity across different sub-populations is depicted in Fig. 4c, where a robust correlation is observed between

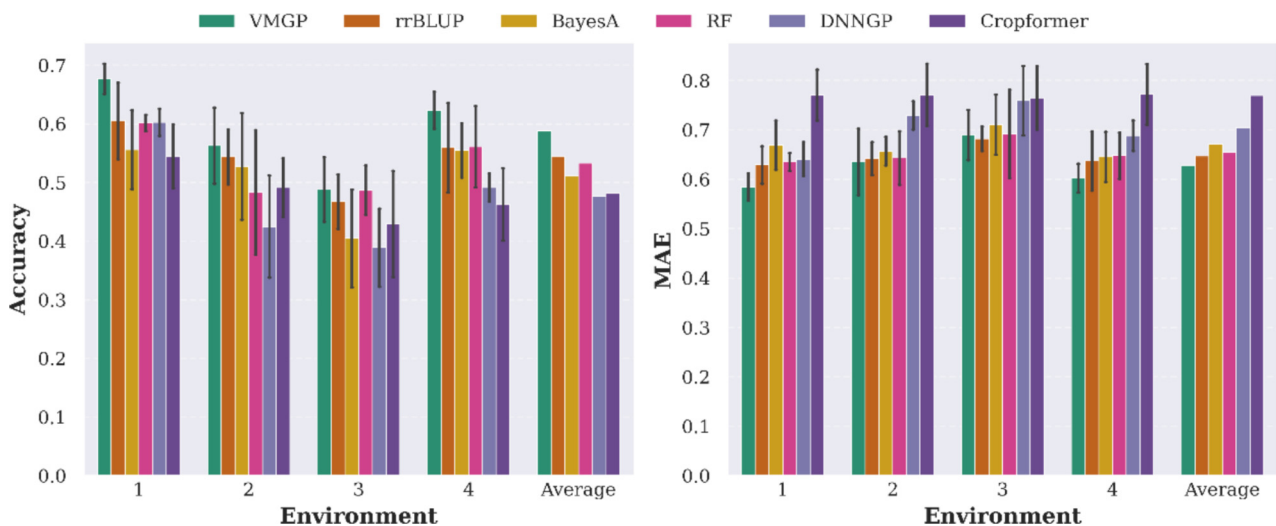


Fig. 3. Multi-environment yield prediction accuracy and prediction error of methods built using four different models on the wheat599 dataset. VMGP, variational auto-encoder based multi-task genomic prediction; rrBLUP, ridge regression-best linear unbiased prediction; RF, random forest; DNNGP, deep neural network for genomic prediction.

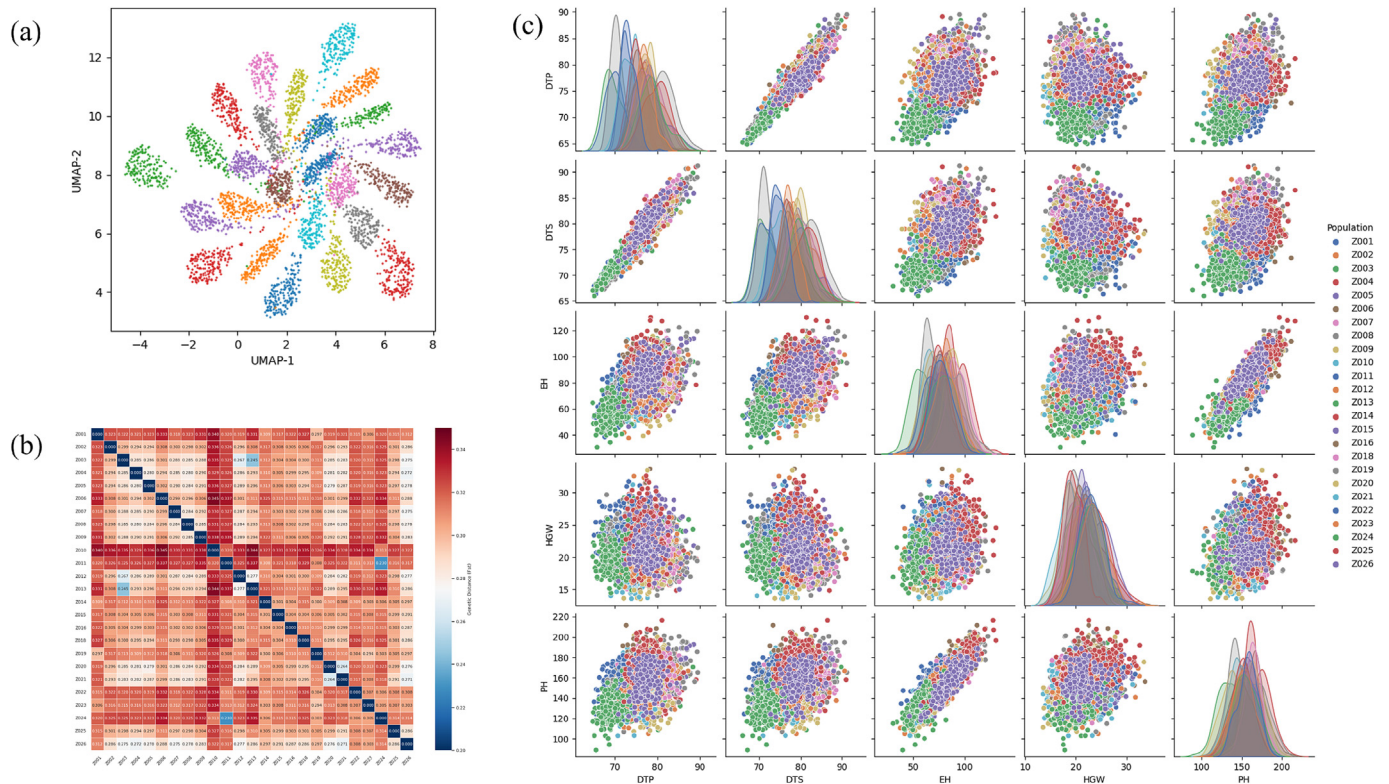


Fig. 4. (a) Two-dimensional visualization of genomic data for each sub-population of the maize4505 dataset. (b) Pairwise fixation index (Fst) analysis of genetic divergence among sub-populations on the maize4505 Dataset. (c) The distribution of phenotypes and correlation between phenotypic traits in each sub-population of the maize4505 dataset.

DTP and DTS, a significant correlation between EH and PH, and a notable absence of correlation between HGW and the remaining traits.

Our proposed VMGP method leverages the power of self-supervised training of genomic data to distill potential genomic insights and employs multi-task learning to unravel the interplay between multiple traits. This approach is designed to delineate the genotype-phenotype relationships without reliance on the population's internal structure and distribution characteristics, transcending the limitations of traditional GS methods that are often tailored to specific traits and populations. To rigorously establish the efficacy of VMGP, we executed a 5-fold cross-validation experiment across the sub-populations within the maize4505 dataset. The dataset was randomly partitioned into five folds based on sub-population criteria, with the stipulation that samples from any given sub-population were solely present in one designated fold. This approach ensured the exclusivity of sub-population representation within each fold. In each iteration of the cross-validation, four folds were allocated for model training, while the remaining fold was sequestered for evaluation. This strategic division guaranteed that the sub-population present in the test set was absent from the training data, thereby enhancing the robustness and authenticity of our cross-population prediction analysis.

In the realm of prediction accuracy, VMGP secured the apex with a marked superiority over comparative methods across three phenotypic traits: DTP, EH, and PH. However, for DTS, Cropformer claimed the highest Accuracy, and for HGW, rrBLUP claimed the highest Accuracy (Fig. 5). Within the same dataset, VMGP exhibited a marginally lower accuracy of 1.3 % in multi-phenotype prediction compared to rrBLUP, yet in cross-population prediction, VMGP's average accuracy surpassed rrBLUP's by a substantial 7.7 %. Particularly notable were its marked improvements in developmental DTP and PH predictions, where Accuracy increased by 14.7 % and 12.8 %, respectively. When juxtaposing cross-population and multi-phenotype prediction, the excess Accuracy of VMGP over RF and DNNGP escalated from 10.8 % and 4.4 % to 37.0 %

and 11.1 %, respectively. Cropformer also demonstrated enhanced cross-population prediction capabilities with 3.0 % improvement in Accuracy compared with rrBLUP, through its multi-head self-attention mechanism. The prediction accuracy and stability of VMGP underscore its proficiency in surmounting the intrinsic structural and distributional constraints of crop populations, effectively exploring the genotype-phenotype relationship, and facilitating cross-population phenotype prediction.

Regarding prediction error, VMGP and DNNGP jointly achieved the most exemplary average performance. VMGP demonstrated the lowest MAE in DTP and EH, while DNNGP outperformed in DTS and PH, and rrBLUP in HGW. However, the discrepancies in prediction error among the various methods (except Cropformer) were not pronounced, with the difference between the best and worst methods being less than 2.8 %. Both VMGP and DNNGP, alongside RF, adopted mean squared error (MSE) as their optimization criterion, yet their prediction accuracies varied significantly, suggesting that different models possess distinct capacities for comprehending the genotype-phenotype relationship.

Consistent with expectations, the prediction accuracy of VMGP-S and VMGP-T was modestly inferior to that of VMGP, with reductions of 4.1 % and 7.5 %, respectively (Fig. S5). When comparing VMGP with VMGP-S, VMGP demonstrated higher Accuracy in predicting four phenotypic traits, DTP, DTS, EH, and PH, but a slightly lower Accuracy in HGW. This may correlate with the observed lack of significant correlation between HGW and other traits in Fig. 4c, and the model's inability to capitalize on data from other traits for optimization.

To systematically evaluate the impact of genetic divergence on cross-population prediction, we implemented a validation framework using the VMGP model trained on sub-populations 2001–2004. Predictive performance was assessed across three sub-populations with different genetic distance: Z010 is the most distant population, with Fst ranging from 0.329 to 0.340 (mean = 0.335); Z026 is the closest one,

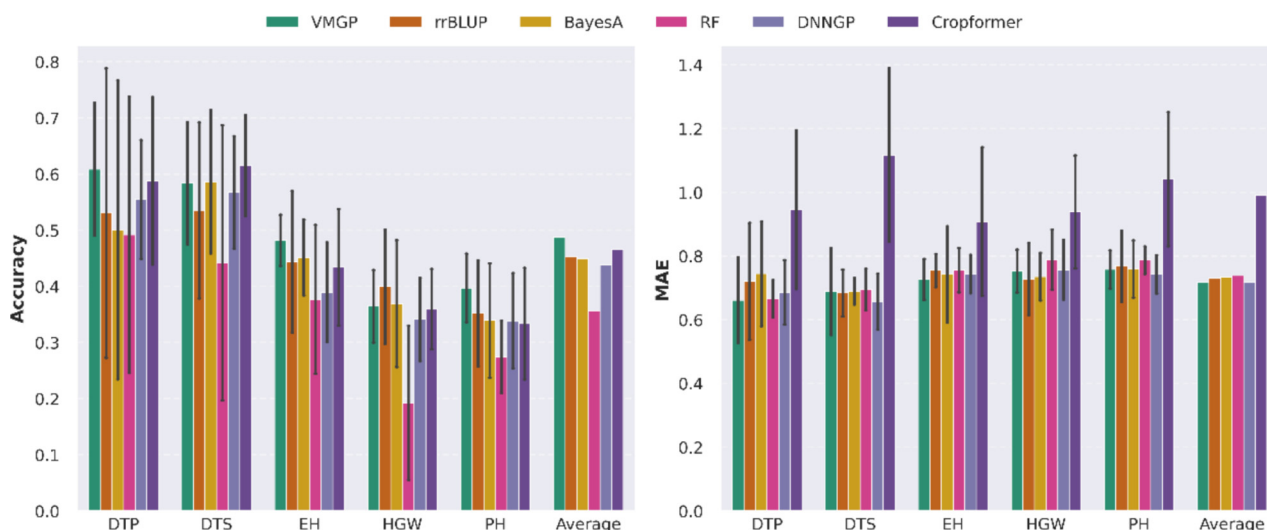


Fig. 5. Cross-population prediction accuracy and prediction error of methods built using four different models on the maize4505 dataset. VMGP, variational auto-encoder based multi-task genomic prediction; rrBLUP, ridge regression-best linear unbiased prediction; RF, random forest; DNNGP, deep neural network for genomic prediction. DTP, days to pollen; DTS, days to silking; EH, ear height; HGW, hundred grain weight; PH, and plant height.

with F_{st} ranging from 0.272 to 0.312 (mean = 0.286); 2014 is an intermediate one, with F_{st} ranging from 0.309 to 0.317 (mean = 0.312). As shown in Fig. S6, the average Accuracy generally decreased with increasing genetic distance (highest in 2026, followed by 2014 and 2010). However, the highest Accuracy of EH and PH is in 2010 while HGW is in 2014. These inconsistencies demonstrated that genetic distance alone cannot fully explain cross-population prediction efficacy, warranting future exploration of detailed genetic architecture factors.

3.4. Model explanation and SNP pre-selection

The VMGP model's capacity for SNP pre-selection, harnessing the power of model explainability, was investigated using SHapley Additive exPlanations (SHAP) (Lundberg and Lee, 2017; Lundberg et al., 2020; Tang et al., 2023), an established methodology for demystifying the internal mechanisms of machine learning models. The SHAP approach is anchored in the estimation of an individual feature's influence on the model's output, treated as a linear function while controlling for other features, with the resulting coefficient known as the Shapley value.

We randomly sampled 10 % of the maize4505 dataset and, post-training with VMGP, computed the Shapley values of each trait for SNP pre-selection. These Shapley values were cumulatively aggregated for SNPs, guiding the strategic selection of SNPs based on their collective significance. Subsequently, a cross-population 5-fold cross-validation was conducted on the residual 90 % of the dataset, employing these pre-selected SNPs as dictated by the model's explanatory insights.

Fig. 6 presents the Shapley value heatmap for the premier 50 SNPs, encapsulating both aggregated and trait-specific values. A robust correlation is evident between the Shapley values for DTP and DTS, as well as EH and PH, echoing the phenotypic correlations observed in Fig. 4c. This alignment suggests that the VMGP model adeptly captures the interplay among multiple traits.

To affirm the potency of the features delineated by the VMGP model, we undertook a comprehensive assessment encompassing 13 discrete feature sets. This spectrum began with sets containing 1000, 2500, 5000, and 10,000 SNPs, and progressively expanded to encompass larger sets, incrementing in blocks of 10,000 SNPs, culminating in an extensive set comprising 100,000 SNPs. Each feature set was carefully selected, predicated on the Shapley values that emerged from our preliminary selection phase. The set of 100,000 SNPs was approximately deemed as the entire set and served as a benchmark for evaluating the performance of the various subsets.

Fig. 7 illustrates the predictive accuracy and error of the VMGP algorithm across this spectrum of pre-selected SNP sets. Remarkably, VMGP, utilizing the top 1000 SNPs, outperformed comparison methods trained on the full SNP set. With 2500 SNPs selected, the predictive performance was comparable to VMGP trained on all SNPs. Most notably, the selection of 10,000 SNPs maximized the predictive performance, with an 11.3 % increase in accuracy and a 6.0 % reduction in error compared to the VMGP method using the entire SNP set. These results highlight the effectiveness of the VMGP model's explainability analysis in identifying key SNPs associated with phenotypic traits, thus enhancing genotype-phenotype predictions.

In contrast, if a subset of SNPs is selected randomly for cross-population prediction, the expected patterns do not manifest, and VMGP's predictive performance may not excel over that achieved using the complete SNP set (Fig. S7). This divergence is attributed to the potential for random SNP selection to miss critical genomic variations contributing to phenotypic variation.

The SNP pre-selection strategy, guided by the VMGP model, can distill a curated subset of SNPs that are tightly correlated with the target phenotypes. By employing targeted sequencing for genotyping (Shirasawa et al., 2016; Rasheed et al., 2017; Yan et al., 2021), the approach not only significantly reduces genotyping costs for large-scale breeding populations but also concurrently enhances the performance of genotype-phenotype predictions.

To validate the biological interpretability of our model, we conducted systematic cross-validation between machine learning-derived feature importance (the Shapley values) and classical QTL mapping results. The QTL analysis was performed using IciMapping v4.2 with the following parameters: a scanning step of 1 cM, a probability in stepwise regression of 0.001, and a Logarithm of Odds (LOD) threshold determined by 1000 permutation tests ($P < 0.05$). Background markers were selected to control residual genetic effects, and significant QTLs were identified based on a minimum LOD score of 2.5. Additive effects, along with phenotypic variance explained (R^2), were estimated for each detected QTL.

QTL mapping of DTP in the 2016 population ($n = 191$ samples) revealed 7 genome-wide significant QTLs distributed across 7 chromosomes. Fig. 8 displayed the LOD values against physical positions, with empirical genome-wide significance threshold ($LOD = 2.5$) marked by a dashed red line. The proposed model explanation revealed functional convergence, with two top SHAP-ranked SNPs (chr2_22590965 and chr7_180135838) co-localizing with these major DTP-QTLs. Our integrated analysis revealed 639 genome-wide significant QTLs ($P < 0.05$)

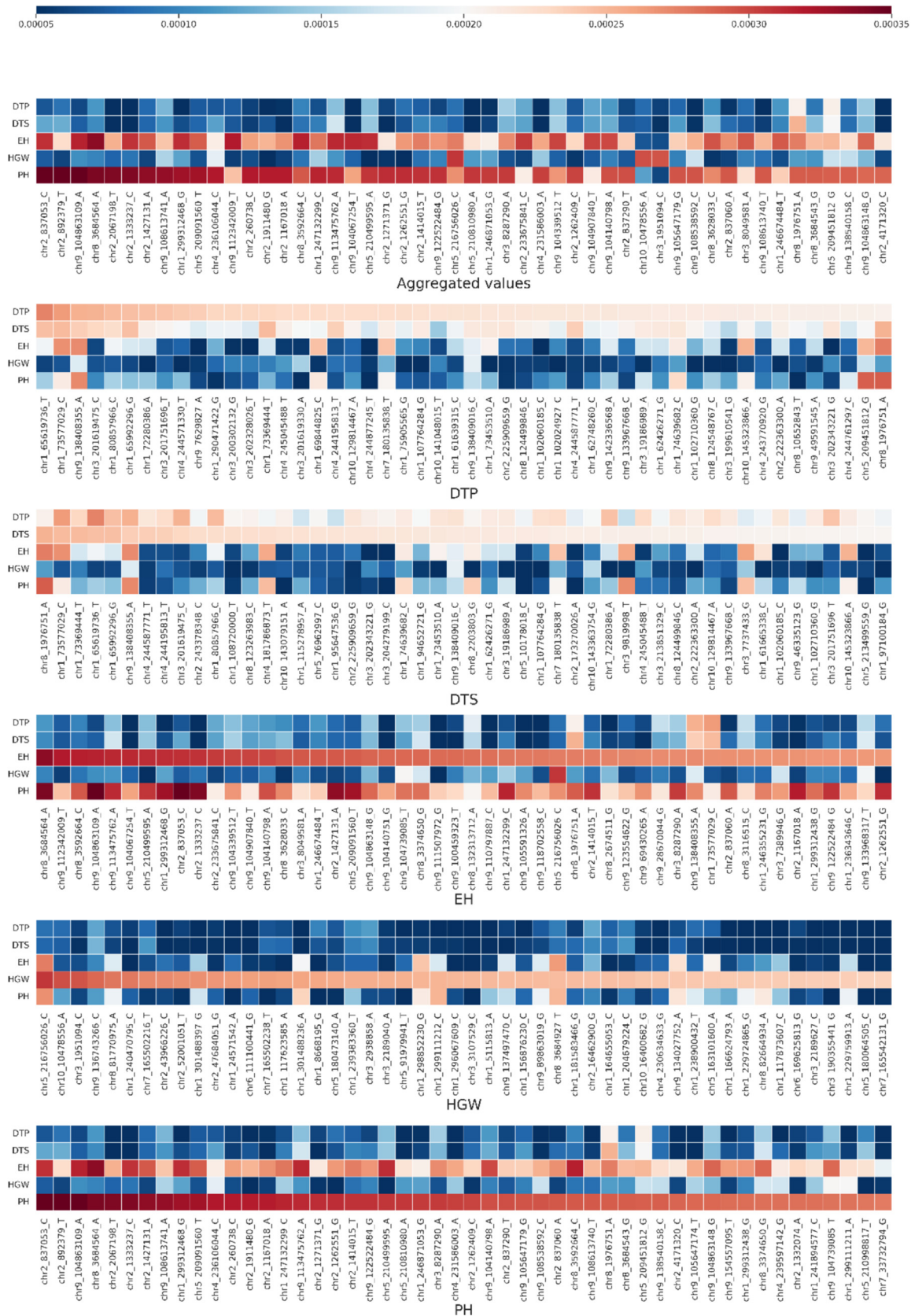


Fig. 6. The Shapley value heatmap of the top 50 SNPs for aggregated values and individual traits. DTP, days to pollen; DTS, days to silking; EH, ear height; HGW, hundred grain weight; PH, and plant height.

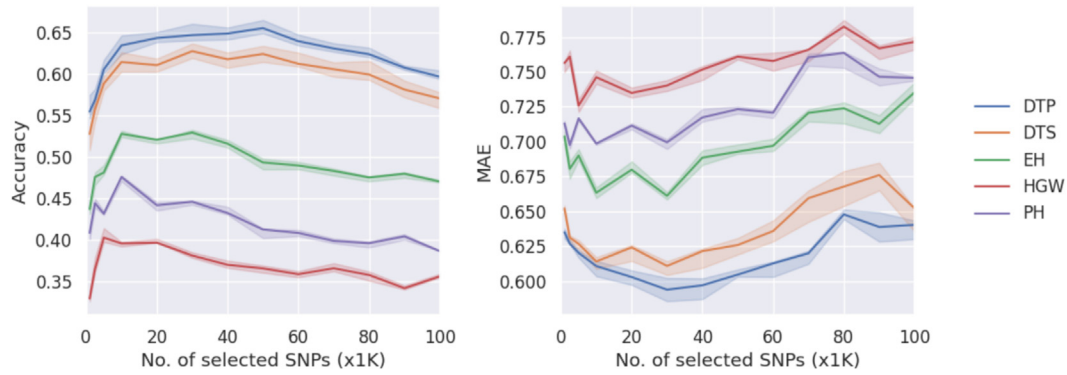


Fig. 7. Cross-population prediction accuracy and prediction error of different SNP numbers pre-selected by VMGP on the maize4505 dataset. VMGP, variational auto-encoder based multi-task genomic prediction. DTP, days to pollen; DTS, days to silking; EH, ear height; HGW, hundred grain weight; PH, and plant height.

across five agronomic traits (Table S2), and 53.4 % (103/193) of top SHAP-ranked SNPs (top 50 per trait) showed positional concordance with QTL confidence intervals (Table S3).

3.5. Hyperparameter selection

This study introduces a parameter-invariant architecture designed for multi-scenario breeding applications, prioritizing structural and prediction accuracy stability. The hyperparameter optimization protocol comprises two phases: structural parameters and loss function configuration.

A comprehensive grid search of structural parameters was conducted across $d1, d2, d3$ and l , with [1024, 2048, 4096], [256, 512, 1024], [64, 128, 256], and [16, 32, 64] as the corresponding range, respectively. As shown in Fig. S8, varying these structural parameters induced insignificant prediction accuracy fluctuations using the wheat599 dataset. The prediction accuracy (mean = 0.578 ± 0.0064) demonstrates statistically insignificant sensitivity to tested hyperparameters across 81 configurations, with all results falling within a narrow band of 0.561–0.592. This parameter-robust behavior remained consistent under 5-fold cross-validation. The observed insensitivity to structural configurations motivated our consistent implementation.

The VMGP framework integrates self-supervised genomic reconstruction with multi-trait genotype-to-phenotype prediction through a unified neural architecture, where the latter is the core task of GS while the former is an auxiliary one. A task-balancing parameter α in the composite loss function modulates the relative contributions of these dual objectives. This multi-task design leverages shared latent representations to disentangle low-dimensional biological embeddings from high-dimensional SNP data, thereby mitigating the CoD problem in GS. Fig. S9 shows the prediction accuracy towards different α settings in the rice299 dataset, where CoD issues are most prominent. When $\alpha = 1$, VMGP did not fit any phenotype data, and as expected, the prediction accuracy of each trait was almost 0. When $\alpha = 0$, VMGP was a pure phenotype prediction method, achieving baseline accuracy 0.351. When α is between 0 and 1, it brought an average accuracy improvement of 8.7 % with the assistance of genomic reconstruction task comparing to the baseline. Our conservative choice $\alpha = 0.5$ balances stability and performance, delivering 9.8 % improvement.

Furthermore, Fig. S10 demonstrates the convergence dynamics of three loss functions (genomic reconstruction, phenotype prediction, and total loss) across 500 training epochs on the wheat599 dataset. The optimization process exhibits a characteristic biphasic pattern:

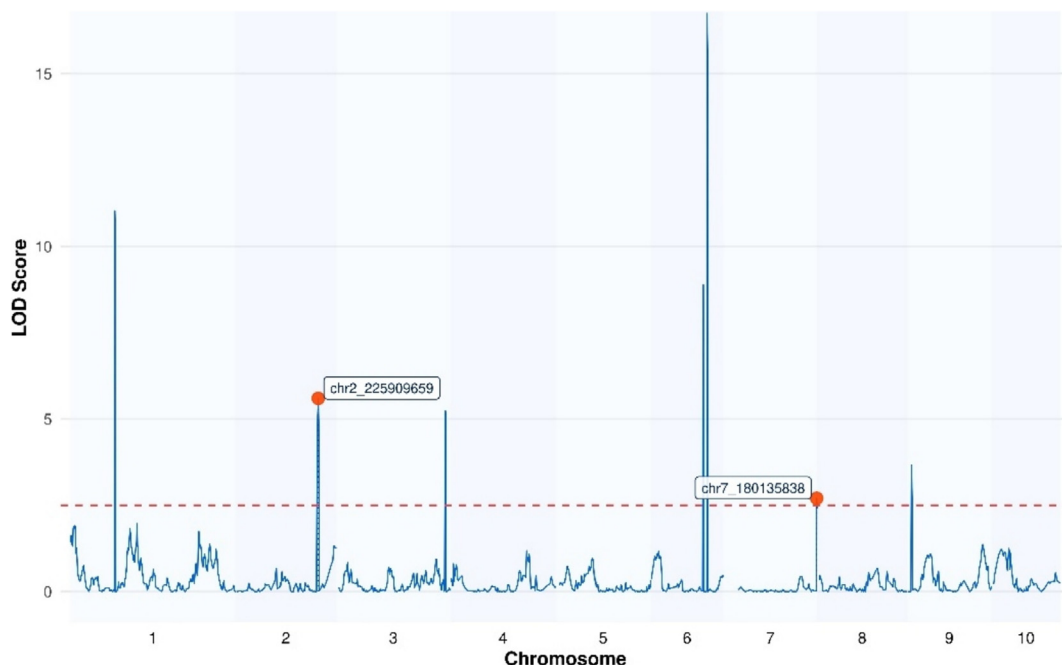


Fig. 8. QTL mapping of DTP in the Z016 population co-localizing with two top SHAP-ranked SNPs of the maize4505 dataset. QTL, quantitative trait loci; DTP, days to pollen.

rapid error reduction during the first 50 epochs followed by progressive stabilization, reaching convergence plateaus beyond epoch 200.

4. Discussion

We have showcased that the VMGP method adeptly manages GS tasks encompassing multi-phenotype prediction, multi-environment prediction, and cross-population prediction. This efficacy is attributed to the synergistic training mechanisms that integrate genomic compression and reconstruction tasks with multiple prediction tasks. Notably, for traits with low heritability, such as the yield trait in the rice299 dataset, VMGP achieves a minimum prediction accuracy improvement of 19.5 % over comparative methods. This enhancement stems from the effective excavation of potential relationships within genotype data, facilitated by the VAE framework, allowing for a more holistic comprehension of the genetic architecture.

VMGP's superiority is particularly evident in multi-environment prediction and correlated trait prediction (e.g., DTP and DTS, EH and PH in the maize4505 dataset). Its performance is primarily due to the effective capture of task coupling relationships through joint training of multiple prediction tasks. This approach fosters a more integrated understanding of the genetic architecture underpinning these traits and their interactions across various environments, thereby enhancing predictive performance. Moreover, the joint training serves as an efficacious strategy to counteract the CoD problem, by extracting commonalities across tasks through a shared network, thus augmenting the model's predictive prowess. This technique empowers the model to adeptly navigate the challenges inherent in high-dimensional genomic data, bolstering its generalization capacity and resilience to noise.

Cross-population GS, characterized by inconsistent distributions between training and test data, presents a complex scenario that often stymies traditional GS methods. In tackling this challenge, VMGP achieves significant accuracy gains over established benchmarks, demonstrating a 7.7 % improvement in accuracy compared to the previous state-of-the-art approach (rrBLUP) and a 4.6 % enhancement over deep learning approach (Cropformer). When integrated with SHAP-guided SNP pre-selection, VMGP's performance differentials expand to 19.9 % and 16.4 % respectively. This exceptional performance underscores VMGP's capability to transcend data distribution barriers between populations, abstracting the genotype-phenotype relationship more effectively and executing GS tasks by harnessing the intrinsic information in genomic data and the insights gleaned from multiple prediction tasks.

Deep learning may not always outperform traditional statistical methods in solving GS problems (Cossa et al., 2017; Ma et al., 2018; Montesinos-López et al., 2021; Nazzicari and Biscarini, 2022; Wang et al., 2023). Even among relatively proficient methods, meticulous parameter selection is imperative. In contrast, our proposed VMGP method employs a unified architecture and parameters across all experiments, attaining the best or near-best predictive performance. This uniform framework, coupled with its simplicity, stable predictive capacity, and open-source code, renders it highly amenable to broad dissemination and application in plant breeding. It is especially beneficial for breeders who prioritize phenotype prediction but may lack in-depth knowledge of deep learning or limited proficiency in parameter tuning. Furthermore, its enhanced cross-population prediction capability suggests the potential to supplant rrBLUP as the foundational algorithm for GS in plant breeding in the era of big data.

The amalgamation of VMGP with the SHAP-based model interpretation method imparts valuable insights into the significance of individual SNPs in predicting various traits. By aggregating Shapley values across multiple traits, VMGP can aid in the pre-selection of SNP subsets, paving the way for potential curtailed genotyping costs. This targeted feature selection can also further refine GS prediction performance. In the aforementioned cross-population task, it was observed that by utilizing merely 2.3 % of preselected SNPs (2500), predictive performance was nearly equivalent to employing all SNPs. This substantial reduction in

the number of SNPs significantly diminishes gene sequencing costs. Moreover, optimal prediction performance was attained when 9.4 % of preselected SNPs (10,000) were engaged, yielding an 11.3 % improvement in prediction accuracy and a 6.0 % reduction in prediction error compared to utilizing all SNPs. Additionally, the selected SNP subset can act as a pre-processing step for delving into functional genes, marking our future research focus.

Ultimately, as a prospective universal genome-wide selection model, numerous avenues remain for ongoing enhancement of this method. Firstly, it is imperative to integrate environmental data to deeply explore the interplay between genes and the environment. The experimental section of this article either treated environmental factors as fixed values or as rudimentary variables, without delving into the specific impacts of environmental factors such as temperature, light, and precipitation on gene expression. We will soon publicly release the MaizeGEP dataset, comprising genotype profiles, multi-environment phenotypic records spanning multiple trial sites (2017–2023), and location-specific daily meteorological matrices. Future work will develop deep learning encoders for environmental covariates, enabling their integration with VMGP to explicitly model genotype-environment interactions, thereby enhancing phenotype prediction accuracy across dynamic agroclimatic conditions. Additionally, transcriptome, proteome, and other information are also pivotal determinants of phenotype. Building upon VMGP, venturing into a GS model for multi-omics data fusion is also a significant endeavor for our subsequent research.

5. Conclusion

This study presents the Variational auto-encoder based Multi-task Genomic Prediction (VMGP) method, which has proven adept at handling the complexities of Genomic Selection (GS) across multi-phenotype, multi-environment, and cross-population scenarios. The method's robustness is underpinned by its synergistic training approach, which seamlessly integrates genomic compression and reconstruction with multiple prediction tasks, thereby enhancing the accuracy of GS.

The VMGP method excels particularly in scenarios with low heritability and in the challenging context of cross-population prediction, surpassing the performance of existing methods. This success is attributed to its ability to effectively uncover and leverage the subtle relationships encoded within genotype data. The method's unified architecture and parameterization, applied consistently across all experiments, have resulted in a model that is not only simple and stable but also highly predictable.

Furthermore, the VMGP method's open-source nature and its capacity for broad application in plant breeding make it an invaluable tool for researchers and breeders alike. Its user-friendly design is particularly beneficial for those prioritizing phenotype prediction without necessitating deep learning expertise or extensive parameter tuning.

As we look to the future, the VMGP method is poised for continuous refinement. The integration of environmental data and the exploration of multi-omics data fusion present promising directions for enhancing the method's predictive power. These advancements will not only deepen our understanding of the interplay between genetic and environmental factors but also contribute to the development of more precise and comprehensive GS models.

CRediT authorship contribution statement

Xiangyu Zhao: Writing – review & editing, Writing – original draft, Software, Methodology, Funding acquisition, Conceptualization. **Fuzhen Sun:** Writing – review & editing, Methodology, Formal analysis. **Jinlong Li:** Writing – review & editing, Software, Data curation. **Dongfeng Zhang:** Writing – review & editing, Visualization, Formal analysis. **Qiusi Zhang:** Writing – review & editing, Visualization,

Software. **Zhongqiang Liu:** Writing – review & editing, Visualization, Funding acquisition, Formal analysis. **Changwei Tan:** Writing – review & editing, Formal analysis. **Hongxiang Ma:** Writing – review & editing, Formal analysis. **Kaiyi Wang:** Writing – review & editing, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ajia.2025.06.007>.

Data availability

The datasets can be downloaded from the references in the Materials and methods. All data supporting the findings of this study are available by the links within the paper and its Supplementary information files. Other relevant data are available upon request. The source code and web-based tool of VMGP are available at <http://user.ebreed.cn:9992/scb/>.

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